In [1]:
```
import numpy as np
import pandas as pd
from io import StringIO
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

data = """User ID,Gender,Age,EstimatedSalary,Purchased
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df = pd.read_csv(StringIO(data))

features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values

for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )
```

```
    model = LogisticRegression()
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)
    if test_score > train_score:
        print("Test {} Train {} Random State {}".format(
            test_score, train_score, i
        ))

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
)

finalModel = LogisticRegression()
finalModel.fit(x_train, y_train)

print(finalModel.score(x_train, y_train))
print(finalModel.score(x_test, y_test))
print(classification_report(label, finalModel.predict(features)))
```

```
Test 0.9 Train 0.840625 Random State 4
Test 0.8625 Train 0.85 Random State 5
Test 0.8625 Train 0.859375 Random State 6
Test 0.8875 Train 0.8375 Random State 7
Test 0.8625 Train 0.8375 Random State 9
Test 0.9 Train 0.840625 Random State 10
Test 0.8625 Train 0.85625 Random State 14
Test 0.85 Train 0.84375 Random State 15
Test 0.8625 Train 0.85625 Random State 16
Test 0.875 Train 0.834375 Random State 18
Test 0.85 Train 0.84375 Random State 19
Test 0.875 Train 0.84375 Random State 20
Test 0.8625 Train 0.834375 Random State 21
Test 0.875 Train 0.840625 Random State 22
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Test 0.85 Train 0.834375 Random State 26
Test 0.85 Train 0.840625 Random State 27
Test 0.8625 Train 0.834375 Random State 30
Test 0.8625 Train 0.85625 Random State 31
Test 0.875 Train 0.853125 Random State 32
Test 0.8625 Train 0.84375 Random State 33
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Test 0.8625 Train 0.853125 Random State 36
Test 0.8875 Train 0.840625 Random State 38
Test 0.875 Train 0.8375 Random State 39
Test 0.8875 Train 0.8375 Random State 42
Test 0.875 Train 0.846875 Random State 46
Test 0.9125 Train 0.83125 Random State 47
Test 0.875 Train 0.83125 Random State 51
Test 0.9 Train 0.84375 Random State 54
Test 0.85 Train 0.84375 Random State 57
Test 0.875 Train 0.84375 Random State 58
Test 0.925 Train 0.8375 Random State 61
Test 0.8875 Train 0.834375 Random State 65
Test 0.8875 Train 0.840625 Random State 68
Test 0.9 Train 0.83125 Random State 72
Test 0.8875 Train 0.8375 Random State 75
Test 0.925 Train 0.825 Random State 76
Test 0.8625 Train 0.840625 Random State 77
Test 0.8625 Train 0.859375 Random State 81
Test 0.875 Train 0.8375 Random State 82
Test 0.8875 Train 0.8375 Random State 83
Test 0.8625 Train 0.853125 Random State 84
Test 0.8625 Train 0.840625 Random State 85
Test 0.8625 Train 0.840625 Random State 87
Test 0.875 Train 0.846875 Random State 88
Test 0.9125 Train 0.8375 Random State 90
Test 0.8625 Train 0.85 Random State 95
Test 0.875 Train 0.85 Random State 99
Test 0.85 Train 0.840625 Random State 101
Test 0.85 Train 0.840625 Random State 102
Test 0.9 Train 0.825 Random State 106
Test 0.8625 Train 0.840625 Random State 107
Test 0.85 Train 0.834375 Random State 109
Test 0.85 Train 0.840625 Random State 111
Test 0.9125 Train 0.840625 Random State 112
Test 0.8625 Train 0.85 Random State 115
Test 0.8625 Train 0.840625 Random State 116
Test 0.875 Train 0.834375 Random State 119
Test 0.9125 Train 0.828125 Random State 120
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Test 0.85 Train 0.846875 Random State 128
Test 0.875 Train 0.85 Random State 130
Test 0.9 Train 0.84375 Random State 133
Test 0.925 Train 0.834375 Random State 134
Test 0.8625 Train 0.85 Random State 135
Test 0.875 Train 0.83125 Random State 138
Test 0.8625 Train 0.85 Random State 141
Test 0.85 Train 0.846875 Random State 143
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Test 0.85 Train 0.84375 Random State 147
Test 0.8625 Train 0.85 Random State 148
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Test 0.925 Train 0.84375 Random State 152
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Test 0.9 Train 0.84375 Random State 154
Test 0.9 Train 0.840625 Random State 155
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Test 0.875 Train 0.828125 Random State 159
Test 0.9 Train 0.83125 Random State 161
Test 0.85 Train 0.8375 Random State 163
Test 0.875 Train 0.83125 Random State 164
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Test 0.8625 Train 0.85 Random State 195
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Test 0.8625 Train 0.8375 Random State 197
Test 0.875 Train 0.840625 Random State 198
Test 0.8875 Train 0.8375 Random State 199
Test 0.8875 Train 0.84375 Random State 200
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Test 0.8625 Train 0.840625 Random State 203
Test 0.8875 Train 0.83125 Random State 206
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Test 0.85 Train 0.84375 Random State 212
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Test 0.875 Train 0.83125 Random State 217
Test 0.9625 Train 0.81875 Random State 220
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Test 0.9 Train 0.84375 Random State 223
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Test 0.8875 Train 0.85 Random State 242
Test 0.8875 Train 0.825 Random State 243
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Test 0.875 Train 0.840625 Random State 245
Test 0.875 Train 0.846875 Random State 246
Test 0.8625 Train 0.859375 Random State 247
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Test 0.8625 Train 0.85 Random State 250
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Test 0.8875 Train 0.84375 Random State 252
Test 0.8625 Train 0.846875 Random State 255
Test 0.9 Train 0.840625 Random State 257
Test 0.8625 Train 0.85625 Random State 260
Test 0.8625 Train 0.840625 Random State 266
Test 0.8625 Train 0.8375 Random State 268
Test 0.875 Train 0.840625 Random State 275
Test 0.8625 Train 0.85 Random State 276
Test 0.925 Train 0.8375 Random State 277
Test 0.875 Train 0.846875 Random State 282
Test 0.85 Train 0.846875 Random State 283
Test 0.85 Train 0.84375 Random State 285
Test 0.9125 Train 0.834375 Random State 286
Test 0.85 Train 0.840625 Random State 290
Test 0.85 Train 0.840625 Random State 291
Test 0.85 Train 0.846875 Random State 292
Test 0.8625 Train 0.8375 Random State 294
Test 0.8875 Train 0.828125 Random State 297
Test 0.8625 Train 0.834375 Random State 300
Test 0.8625 Train 0.85 Random State 301
Test 0.8875 Train 0.85 Random State 302
Test 0.875 Train 0.846875 Random State 303
Test 0.8625 Train 0.834375 Random State 305
Test 0.9125 Train 0.8375 Random State 306
Test 0.875 Train 0.846875 Random State 308
Test 0.9 Train 0.84375 Random State 311
Test 0.8625 Train 0.834375 Random State 313
Test 0.9125 Train 0.834375 Random State 314
Test 0.875 Train 0.8375 Random State 315
Test 0.9 Train 0.846875 Random State 317
Test 0.9125 Train 0.821875 Random State 319
Test 0.8625 Train 0.85 Random State 321
Test 0.9125 Train 0.828125 Random State 322
Test 0.85 Train 0.846875 Random State 328
Test 0.85 Train 0.8375 Random State 332
Test 0.8875 Train 0.853125 Random State 336
Test 0.85 Train 0.8375 Random State 337
Test 0.875 Train 0.840625 Random State 343
Test 0.8625 Train 0.84375 Random State 346
Test 0.8875 Train 0.83125 Random State 351
Test 0.8625 Train 0.85 Random State 352
Test 0.95 Train 0.81875 Random State 354
Test 0.8625 Train 0.85 Random State 356
Test 0.9125 Train 0.840625 Random State 357
Test 0.8625 Train 0.8375 Random State 358
Test 0.85 Train 0.840625 Random State 362
Test 0.9 Train 0.84375 Random State 363
Test 0.8625 Train 0.853125 Random State 364
Test 0.9375 Train 0.821875 Random State 366
Test 0.9125 Train 0.840625 Random State 369
Test 0.8625 Train 0.853125 Random State 371
Test 0.925 Train 0.834375 Random State 376
Test 0.9125 Train 0.828125 Random State 377
```

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Test 0.8875 Train 0.85 Random State 378
Test 0.8875 Train 0.85 Random State 379
Test 0.8625 Train 0.840625 Random State 382
Test 0.8625 Train 0.859375 Random State 386
Test 0.85 Train 0.8375 Random State 387
Test 0.875 Train 0.828125 Random State 388
Test 0.85 Train 0.84375 Random State 394
Test 0.8625 Train 0.8375 Random State 395
Test 0.9 Train 0.84375 Random State 397
Test 0.8625 Train 0.84375 Random State 400
0.865625
0.8125
              precision    recall  f1-score   support

           0       0.86      0.92      0.89       257
           1       0.83      0.74      0.79       143

    accuracy                           0.85       400
   macro avg       0.85      0.83      0.84       400
weighted avg       0.85      0.85      0.85       400
```

In [ ]: